Unit 2

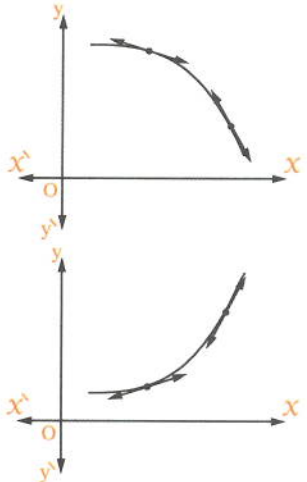
Lesson Four

Convexity of curves and inflection points

⊙ A continuous part of a curve is said to be :

1. Convex upwards "concave downwards"
   If the curve lies
   below all its tangents.

2. Convex downwards "concave upwards".
   If the curve lies
   above all its tangents.



Definition

⊙ If f is a differentiable function on the interval $]a , b[$, then the curve of the function f

1. Convex downwards — If f' is increasing on $]a , b[$
2. Convex upwards — If f' is decreasing on $]a , b[$

Note that

If f' is increasing , then its derivative is positive — i.e. $f''(x) > 0$

and if f' is decreasing , then its derivative is negative — i.e. $f''(x) < 0$

and so we determine the intervals of convex upwards and convex downwards to the curve of the function f during the study of the increasing and decreasing of the first derivative f' , then we can deduce the following theorem.

Theorem

"The second derivative test for convexity of curves"

Let f be a differentiable function twice on the interval $]a , b[$

1. If $f''(x) > 0$ for all the values of $x \in]a , b[$, then the curve of f is convex downwards on the interval $]a , b[$
2. If $f''(x) < 0$ for all the values of $x \in]a , b[$, then the curve of f is convex upwards on the interval $]a , b[$

Definition of the inflection point

If f is a continuous function on the open interval $]a , b[$, $c \in]a , b[$ and the curve of the function has a tangent at point C $(c , f(c))$, then this point is called the inflection point to the curve of the function f if the convexity of the curve of the function changes at this point from being convex downward to convex upwards or vice versa.

From the previous definition we deduce that :

The point C $(c , f(c))$ is an inflection point of a curve of a function f if the following is satisfied.

1. The curve of the function f is continuous at C

2. We can draw one tangent to the curve of the function at C

Where : (1) $\grave{f}(c) \in \mathbb{R}$ **i.e.** The tangent is inclined or horizontal.

(2) $\grave{f}(c) = \pm \infty$ (denominator of $\grave{f}$ at the point C = zero)

i.e. The tangent is vertical.

3. The sign of $\acute{f}(x)$ changes before and after the point C

i.e. $[\acute{f}(c) = 0$ or not exist$]$

and the following figures illustrate that :

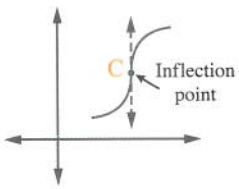


Fig. (1)

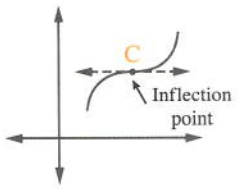


Fig. (2)

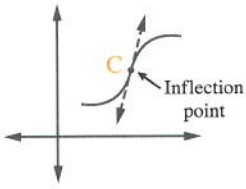


Fig. (3)

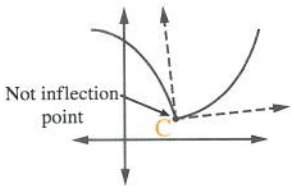


Fig. (4)

* There are inflection points in fig. (1) , (2) , (3) because they are curves of a continuous functions have a tangent at C and the direction of the curve convexity changes before and after the point C

* There are not inflection point in fig. (4) because the absence of a unique tangent at the point C

Important confirmation

The inflection point at $x = a$ must be belong to the domain of the function f

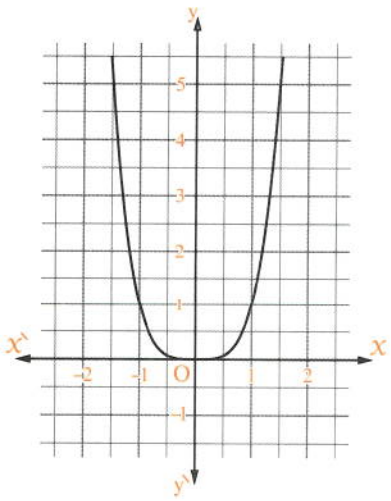
i.e. $f(a)$ is defined.

Remarks

1. The inflection points are the points which separated between convex upwards and convex downwards regions.

2. The tangent at the inflection point intersects the curve of the function.

3. [If $(c , f(c))$ is an inflection point of the curve of the function , then $f''(c)$ equals zero if it exists] but the opposite is not true because [If $f''(c)$ exist and equals zero , then this point is not inflection point unless the sign of $f''(x)$ changes before and after the point C]



For example : $f(x) = x^4$

we find that : $f'(x) = 4x^3$, $f''(x) = 12x^2 > 0$

for all $x \in \mathbb{R} - \{0\}$

$\therefore$ The curve always convex downwards

$\therefore$ There is not inflection point to this curve inspite of $f''(0) = 0$

4. The critical point of the function f is the point at which $f'(x) = 0$ or not exist and if the sign of $f'(x)$ changes around this point , then it is a local maximum or minimum.

Also we find that :

The critical point of the function f' is the point at which $f''(x) = 0$ or not exist and if the sign of $f''(x)$ changes around this point , then it is an inflection point for the function f

5. The inflection points of a function f which is differentiable twice is a local maximum or minimum of the function f'

6. If the point $(c , h) \in$ the curve of a function which is differentiable twice and

(a) (c , h) is an inflection point , then $f''(c) = 0$, $f(c) = h$

(b) (c , h) is a local maximum or a local minimum or critical point , then $f'(c) = 0$, $f(c) = h$

7. The polynomial function is differentiable at all of the points of its domain , therefore it has a tangent at any point on it.

Steps to study the convexity interval and the inflection points :

1. Find $f''(x)$, then find the values of x which make $f''(x) = 0$ or not exist.
2. Determine the sign of $f''(x)$ to determine the intervals over which the function is convex upwards where $[f''(x) < 0]$ and the intervals over which the function is convex downwards where $[f''(x) > 0]$
3. Determine the inflection points from the points that we obtain where the sign of $f''(x)$ changes on right and left each point and if the sign of $f''(x)$ does not change about any of these points , then it is not inflection point.

Example 1

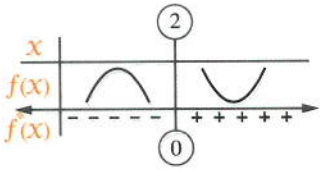
Determine the intervals of the convexity upwards and downwards and the inflection points of the curve of the function f : $f(x) = \frac{1}{3}x^3 - 2x^2 + 4x - 5$

Solution

$\because f(x) = \frac{1}{3}x^3 - 2x^2 + 4x - 5$

$\therefore f'(x) = x^2 - 4x + 4$, $f''(x) = 2x - 4$

put $f''(x) = 0$ $\quad \therefore 2x - 4 = 0$ $\quad \therefore x = 2$



$\therefore$ The curve is convex upwards on $]-\infty , 2[$

, the curve is convex downwards on $]2 , \infty[$

, $\because$ the sign of $f''(x)$ changes before and after $x = 2$

$\therefore$ There is an inflection point at $x = 2$

, $\because f(2) = \frac{-7}{3}$ $\quad \therefore$ The inflection point is $\left(2 , \frac{-7}{3}\right)$

Example 2

Investigate the convexity of the curve of the function f : $f(x) = x^2 - 2$ and find the inflection points (If found).

Solution

$\because f(x) = x^2 - 2$ $\quad \therefore f'(x) = 2x$, $f''(x) = 2$

$\therefore f''(x) > 0$ for all $x \in \mathbb{R}$

$\therefore$ The curve of the function is always convex downwards.

, $\because$ the curve does not change its convexity at any point.

$\therefore$ There is no inflection points for this function.

Example 3

Investigate the convexity of the curve of the function $f : f(x) = x^4 + x^3 - 3x^2 + 1$, and find the inflection points.

Solution

$\because f(x) = x^4 + x^3 - 3x^2 + 1$

$\therefore \grave{f}(x) = 4x^3 + 3x^2 - 6x$

, $\acute{f}(x) = 12x^2 + 6x - 6$

put $\acute{f}(x) = 0$

$\therefore 12x^2 + 6x - 6 = 0$

$\therefore 2x^2 + x - 1 = 0$

$\therefore (2x - 1)(x + 1) = 0$

$\therefore x = \frac{1}{2}$ or $x = -1$

$\therefore$ The curve is convex upwards on $]-1 , \frac{1}{2}[$

, the curve is convex downwards

on $]-\infty , -1[$, $]\frac{1}{2} , \infty[$

, $\because$ the sign of $\acute{f}(x)$ changes before

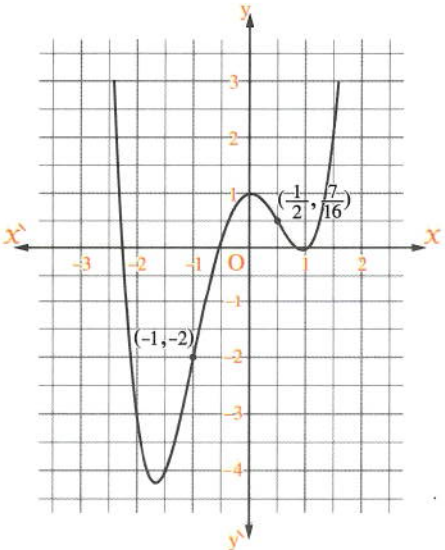
and after each of $x = -1$, $x = \frac{1}{2}$

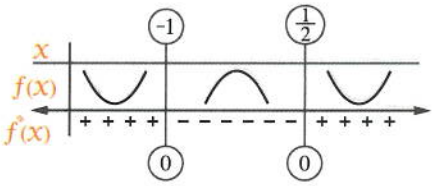
$\therefore$ There are inflection points at $x = -1$, $x = \frac{1}{2}$

, $f(-1) = -2$, $f\left(\frac{1}{2}\right) = \frac{7}{16}$

$\therefore$ The inflection points are $(-1 , -2)$, $\left(\frac{1}{2} , \frac{7}{16}\right)$

For reading only





Example 4

Investigate the convexity of the curve of the function $f : f(x) = \sqrt[3]{x-3}$ and find the inflection points.

Solution

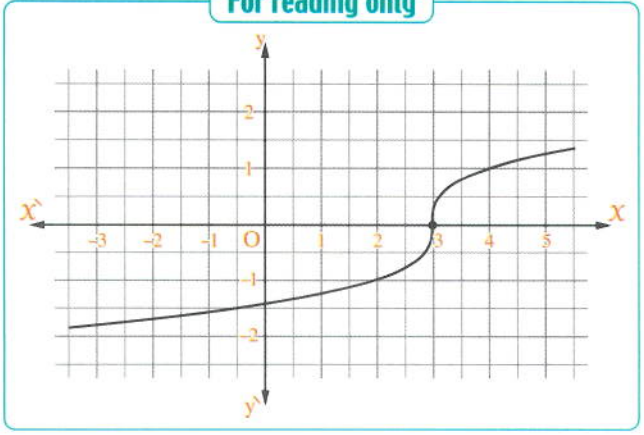
$\because f(x) = \sqrt[3]{x-3} = (x-3)^{\frac{1}{3}}$

$\therefore \grave{f}(x) = \frac{1}{3}(x-3)^{\frac{-2}{3}}$

, $\acute{f}(x) = \frac{-2}{9}(x-3)^{\frac{-5}{3}}$

$= \frac{-2}{9} \times \frac{1}{\sqrt[3]{(x-3)^5}}$

For reading only



$f''(X)$ is undefined when $X - 3 = 0$

$\therefore X = 3$

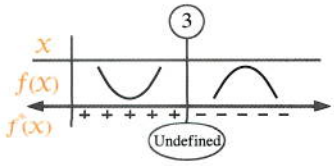
$\therefore$ The curve is convex upwards on $]3 , \infty[$

, the curve is convex downwards on $]-\infty , 3[$

, $\because$ the sign of $f''(X)$ changes before and after $X = 3$

There is inflection point at $X = 3$, $f(3) =$ zero

$\therefore$ The inflection point is $(3 , 0)$



Notice that

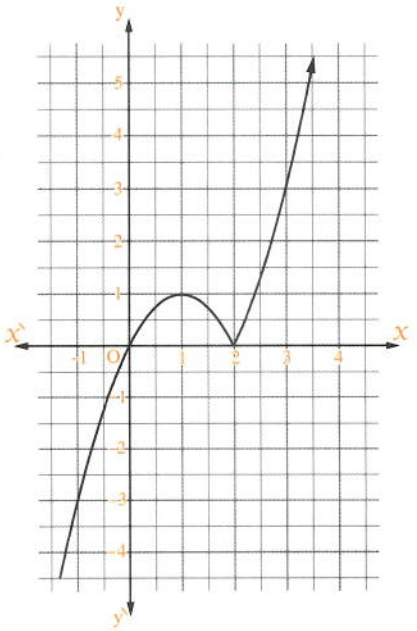
The slope of the tangent to the continuous curve at $X = 3$ is undefined.

i.e. There is a vertical tangent at $X = 3$

Example 5

Investigate the convexity of the curve of the function $f : f(X) = X|X-2|$ and find the inflection point (If found)

For reading only



Solution

$$f(X) = \begin{cases} X(X-2) & , X \geq 2 \\ -X(X-2) & , X < 2 \end{cases}$$

$$= \begin{cases} X^2 - 2X & , X \geq 2 \\ -X^2 + 2X & , X < 2 \end{cases}$$

, $\because f'(2^+) \neq f'(2^-)$

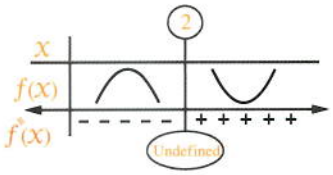
$\therefore$ The function is not differentiable at $X = 2$

$$\therefore f'(X) = \begin{cases} 2X - 2 & , X > 2 \\ \text{not exist} & , X = 2 \\ -2X + 2 & , X < 2 \end{cases}$$

$$, f''(X) = \begin{cases} 2 & , X > 2 \\ \text{not exist} & , X = 2 \\ -2 & , X < 2 \end{cases}$$

$\therefore$ The curve is convex upwards on $]-\infty , 2[$

, the curve is convex downwards on $]2 , \infty[$



Notice that

There is no inflection point at $X = 2$ in spite of the sign of $f''(X)$ changes before and after $X = 2$ because the function is not differentiable at $X = 2$ for $[f'(2^+) \neq f'(2^-)]$

i.e. We can not draw a unique tangent to this function at $X = 2$

Example 6

Find a , b , c and d such that the function $f(x) = a x^3 + b x^2 + c x + d$ satisfies the following conditions

1. Passes through (1 , 2)

2. Has an inflection point at (2 , 1)

3. The tangent to it at its inflection point is horizontal.

Solution

$\because f(x) = a x^3 + b x^2 + c x + d$

$\therefore \grave{f}(x) = 3 a x^2 + 2 b x + c$

$\therefore \grave{\grave{f}}(x) = 6 a x + 2 b$

, $\because$ the curve passes through (1 , 2)

$\therefore a + b + c + d = 2$ (1)

, $\because$ the curve has an inflection point at (2 , 1)

$\therefore f(2) = 1$

$\therefore 8 a + 4 b + 2 c + d = 1$ (2)

, $\grave{\grave{f}}(2) = 0$ $\therefore 12 a + 2 b = 0$

$\therefore 6 a + b = 0$ (3)

$\because$ The tangent at (2 , 1) is horizontal $\therefore \grave{f}(2) = 0$

, $\therefore 12 a + 4 b + c = 0$ (4)

adding (1) , (4) and subtracting (2)

$\therefore 5 a + b = 1$ (5)

subtract (5) from (3) $\therefore a = -1$

, from (5) $\therefore b = 6$

, from (4) $\therefore 12(-1) + 4(6) + c = 0$ $\therefore c = -12$

, from (1) $\therefore d = 9$